

WASKESIU LAKE, SK, Canada

WMO: 714540

Lat: 53.920N

Lon: 106.070W

Elev: 569

StdP: 94.67

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.0	(c) -30.4	(d) -36.8	(e) 0.1	(f) -32.8	(g) -34.0	(h) 0.2	(i) -30.3	(j) 6.9	(k) -9.5	(l) 6.1	(m) -10.1	(n) 1.5	(o) 310	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	27.5	17.6	25.8	16.8	24.2	15.8	19.1	25.2	18.0	23.6	17.1	22.2	2.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	13.0	21.0	16.0	12.2	20.2	15.0	11.4	19.3	56.8	25.4	53.2	23.7	50.2	22.2	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.1	(b) 5.4	(c) 4.8	(d) DB	(e) -35.9	(f) 31.1	(g) 2.5	(h) 1.7	(i) -37.7	(j) 32.3	(k) -39.2	(l) 33.3	(m) -40.6	(n) 34.3	(o) -42.5	(p) 35.5	(q) 23.8
(a) 6.1	(b) 5.4	(c) 4.8	(d) WB	(e) -36.0	(f) 21.2	(g) 2.5	(h) 1.0	(i) -37.8	(j) 21.9	(k) -39.3	(l) 22.5	(m) -40.7	(n) 23.1	(o) -42.5	(p) 23.8	(q) 23.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-15.8	-13.4	-6.6	2.1	9.3	14.6	17.5	16.2	10.9	3.2	-6.4	-13.7
	DBStd	13.27	8.75	7.76	7.79	5.89	4.92	3.63	2.95	3.41	4.41	4.74	6.56	8.07
	HDD10.0	3777	801	654	516	244	74	6	0	1	41	216	491	733
	HDD18.3	6168	1059	888	774	488	282	119	52	83	223	469	741	992
	CDD10.0	703	0	0	0	6	52	145	232	194	69	6	0	0
	CDD18.3	53	0	0	0	0	1	8	25	17	2	0	0	0
	CDH23.3	534	0	0	0	1	30	89	221	163	29	1	0	0
	CDH26.7	89	0	0	0	0	6	15	36	29	3	0	0	0
(14) Wind	WSAvg	2.2	2.1	2.1	2.4	2.5	2.4	2.2	2.1	2.0	2.2	2.5	2.4	2.1
(15) Precipitation	PrecAvg	477	18	14	19	32	42	88	80	58	44	39	22	20
	PrecMax	622	31	24	39	80	126	238	151	116	165	118	57	83
	PrecMin	278	3	2	1	6	2	40	42	15	1	4	1	3
	PrecStd	93	6	7	11	22	31	40	30	28	36	28	18	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	6.5	12.1	21.0	26.8	28.6	29.9	29.7	26.7	20.3	9.7	4.7
		MCWB	2.1	2.1	5.1	9.9	14.4	17.1	19.0	18.7	16.2	12.0	5.0	1.7
	2%	DB	1.7	2.8	8.5	17.2	23.2	25.8	27.5	27.1	23.4	15.5	6.3	1.6
		MCWB	-0.1	-0.1	3.1	8.4	12.2	15.9	18.3	17.7	14.7	9.2	3.0	-0.4
	5%	DB	-0.8	0.1	6.1	14.4	20.8	23.8	25.8	25.1	20.9	12.6	3.6	-0.7
		MCWB	-1.9	-1.5	1.6	6.9	10.9	14.8	17.6	16.7	13.5	7.7	1.3	-1.9
	10%	DB	-3.3	-2.2	3.4	11.5	18.6	21.9	24.1	23.2	18.3	10.2	1.6	-2.9
		MCWB	-4.1	-3.3	0.5	5.2	10.0	13.8	16.6	16.0	12.4	6.4	0.1	-3.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	2.6	5.6	11.0	15.1	19.3	21.2	20.1	17.2	12.4	5.5	2.1
		MCDB	5.0	5.8	11.5	18.6	22.5	26.1	26.9	26.5	23.9	19.3	9.2	4.2
	2%	WB	0.1	0.3	3.5	8.9	13.4	17.4	19.5	18.7	15.5	10.1	3.3	-0.1
		MCDB	1.4	2.5	7.8	16.3	20.8	22.5	25.3	25.1	21.6	14.1	5.6	1.4
	5%	WB	-1.9	-1.5	2.0	7.1	12.0	16.2	18.4	17.7	14.2	8.2	1.6	-1.9
		MCDB	-0.9	0.0	5.3	13.5	18.9	21.2	23.9	23.5	19.4	11.9	3.3	-0.7
	10%	WB	-4.1	-3.3	0.6	5.6	10.7	15.1	17.5	16.7	12.9	6.7	0.2	-3.7
		MCDB	-3.2	-2.2	3.2	11.0	17.3	20.1	22.5	21.8	17.7	9.9	1.5	-2.9
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.5	11.1	11.0	11.9	10.5	10.5	10.3	9.8	7.8	6.6	7.6
		MCDBR	10.4	12.2	12.8	14.9	15.0	13.7	13.0	13.2	14.2	12.6	9.0	9.5
		MCWBR	8.7	9.4	8.0	7.9	6.7	6.0	5.7	5.9	7.0	7.2	6.2	7.9
	5% WB	MCDBR	10.0	11.7	11.9	13.9	13.6	11.2	11.5	11.8	12.8	11.9	8.3	9.1
		MCWBR	8.5	9.2	7.7	7.9	6.6	5.8	5.8	5.9	6.8	7.1	6.1	7.8
(40) Clear-Sky Solar Irradiance	taub		0.230	0.236	0.277	0.317	0.339	0.361	0.364	0.356	0.307	0.288	0.248	0.219
	taud		2.345	2.378	2.357	2.439	2.437	2.407	2.395	2.430	2.579	2.579	2.471	2.356
	Ebn at Noon		777	895	915	917	909	888	877	864	878	824	755	722
	Edh at Noon		53	72	93	99	105	110	109	100	76	60	47	42
(44) All-Sky Solar Radiation	RadAvg		0.91	1.96	3.51	4.80	5.64	5.65	5.75	4.79	3.32	1.89	0.84	0.62
	RadStd		0.09	0.11	0.25	0.40	0.50	0.41	0.40	0.41	0.37	0.30	0.13	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.34	-0.46	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page